

ALF Tutorial – 2.3

Examples and exercises for the *ALF* (Algorithms for Lattice Fermions) package

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The ALF package provides a general code for auxiliary-field Quantum Monte Carlo simulations and default analysis. In this tutorial we show how users from beginners to specialists can profit from ALF. This document is divided in two parts:

Part I. The first, introductory part of the tutorial is based on ALF's python interface – **pyALF** – which greatly simplifies using the code, making it ideal for: *obtaining benchmark* results for established models; *getting started* with QMC and ALF; or just *quickly running* a simulation.

Part II. The second part is independent of the first and aimed at more advanced users who want to simulate their own systems. It guides the user on how to modify the package's Fortran source code and presents the resources implemented to facilitate this task.

This document is intended to be self-contained, but the interested reader should check [ALF's documentation](#), which contains a thorough, systematic description of the package in its 2.3 version (or [here](#) for its latest (development) version).

Part I. Just run it

What follows is a collection of self-explanatory Jupyter notebooks written in Python, each centered on a detailed example followed by a few simple exercises. The notebooks printed below can be found, together with the necessary files and an increasing number of additional notebooks exploring ALF's capabilities, in the [pyALF repository](#), branch ALF-2.3.

Requirements

You can download pyALF from its repository linked above or, from the command line¹:

```
git clone -b ALF-2.3 https://git.physik.uni-wuerzburg.de/ALF/pyALF.git
```

To run the notebooks you need the following installed in your machine:

- Python and its packages SciPy, NumPy, matplotlib, f90nml, h5py, pandas, numba, tkinter, ipywidgets and ipynb
- Jupyter
- the libraries Lapack and Blas
- a Fortran compiler, such as **gfortran** or **ifort**,

where the last two are required by the main package [ALF](#). Also, add pyALF's path to your environment variable PYTHONPATH. In Linux, this can be achieved, e.g., by adding the following line to **.bashrc**:

```
export PYTHONPATH="/local/path/to/pyALF:$PYTHONPATH"
```

¹Avoid folder names containing spaces, they are not supported by the installation scripts.

Python and its packages can be easily installed on a variety of platforms using the Anaconda distribution – check its [installation instructions](#) for your system. Then, from Anaconda, you can issue the command

```
conda install -c anaconda ipython jupyterlab scipy numpy matplotlib h5py pandas  
↪ numba tkinter ipywidgets ipynb f90nml
```

Anaconda is recommended due to its convenience, but the system’s package management (e.g., apt-get) or Python’s own package management, pip3, can be used instead if preferred – see, for instance, SciPy [installation instructions](#).

A Fortran compiler and the libraries needed for ALF can be installed in a Debian-based Linux via

```
sudo apt-get install gfortran liblapack-dev make
```

In MacOS, gfortran can be found at <https://gcc.gnu.org/wiki/GFortranBinaries#MacOS>, where detailed instructions [are available](#). You will need to have Xcode as well as the Apple developer tools installed.

For Windows and other Linuxes and Unixes, please check the Tutorials’ repository [README](#).

Starting

Jupyter notebooks [are run](#) through a Jupyter server² started, e.g., from the command line:

```
jupyter-lab
```

(or, depending on the installation, `jupyter lab`, or `jupyter-notebook` or `jupyter notebook`) which opens the “notebook dashboard” in your default browser, where you can navigate through your file structure to the pyALF directory. There you will find the interface’s core module, `py_alf.py`, some auxiliary files, and notebooks such as the ones included below. Have fun.

Notebooks

1. A minimal ALF run

In this bare-bones example we use the [pyALF](#) interface to run the canonical Hubbard model on a default configuration: a 6×6 square grid, with interaction strength $U = 4$ and inverse temperature $\beta = 5$.

Bellow we go through the steps for performing the simulation and outputting observables.

1. Import `ALF_source` and `Simulation` classes from the `py_alf` python module, which provide the interface with ALF:

```
[1]: from py_alf import ALF_source, Simulation # Interface with ALF
```

2. Create an instance of `ALF_source`, downloading the ALF source code from the [ALF repository](#), if `alf_dir` does not exist. Gets `alf_dir` from environment variable `$ALF_DIR`, or defaults to “./ALF”, if not present:

```
[2]: alf_src = ALF_source(branch='ALF-2.3')
```

```
Checking out branch ALF-2.3
```

```
Your branch is up to date with 'origin/ALF-2.3'.
```

```
Already on 'ALF-2.3'
```

²Note that pyALF can also be used to start a simulation from the command line, without starting a Jupyter server or using a notebook. For instance: `python3.7 Run.py -R -alfdir /home/debian/ALF-1.2/ -config "Intel" -executable_R Hubbard -mpi True` starts a parallel run of the Hubbard model, using ALF compiled with `ifort`. Notice that `Run.py` requires a configuration file `Sims`, which defines the simulation parameters. An entry of `Sims` might read as: `"Model": "Hubbard", "Lattice_type": "Square", "L1": 4, "L2": 4, "NBin": 5, "ham_T": 0.0, "Nsweep": 2000, "Beta": 1.0, "ham_chem": -1.0`

3. Create an instance of `Simulation`, overwriting default parameters as desired:

```
[3]: sim = Simulation(  
    alf_src,  
    "Hubbard",           # Name of Hamiltonian  
    {                    # Dictionary overwriting default parameters  
        "Lattice_type": "Square"  
    },  
    machine='GNU' # Change to "intel", or "PGI" if gfortran is not installed  
)
```

4. Compile ALF. The first time it will also download and compile HDF5, which could take ~ 15 minutes.

```
[4]: sim.compile()
```

```
Compiling ALF...  
Cleaning up Prog/  
Cleaning up Libraries/  
Cleaning up Analysis/  
Compiling Libraries  
  
entanglement_mod.F90:35:2:  
  
    35 | #warning "You are compiling entanglement without MPI. No results  
possible"  
      | 1~~~~~  
Warning: #warning "You are compiling entanglement without MPI. No results  
possible" [-Wcpp]  
ar: creating modules_90.a  
ar: creating libqrref.a  
  
Compiling Analysis  
Compiling Program  
Parsing Hamiltonian parameters  
filename: Hamiltonians/Hamiltonian_Kondo_smod.F90  
filename: Hamiltonians/Hamiltonian_Hubbard_smod.F90  
filename: Hamiltonians/Hamiltonian_Hubbard_Plain_Vanilla_smod.F90  
filename: Hamiltonians/Hamiltonian_tV_smod.F90  
filename: Hamiltonians/Hamiltonian_LRC_smod.F90  
filename: Hamiltonians/Hamiltonian_Z2_Matter_smod.F90  
Compiling program modules  
Link program  
Done.
```

5. Perform the simulation as specified in `sim`:

```
[5]: sim.run()
```

```
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square"  
for Monte Carlo run.  
Create new directory.  
Run /home/jonas/Programs/ALF/Prog/ALF.out  
ALF Copyright (C) 2016 - 2021 The ALF project contributors  
This Program comes with ABSOLUTELY NO WARRANTY; for details see license.GPL  
This is free software, and you are welcome to redistribute it under certain  
conditions.  
No initial configuration
```

6. Perform some simple analysis:

```
[6]: sim.analysis()
```

```
### Analyzing /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square ###  
/home/jonas/Programs/pyALF/Notebooks  
Scalar observables:  
Ener_scal  
Kin_scal  
Part_scal  
Pot_scal  
Histogram observables:  
Equal time observables:  
Den_eq  
Green_eq
```

```

SpinT_eq
SpinXY_eq
SpinZ_eq
Time displaced observables:
Den_tau
Green_tau
SpinT_tau
SpinXY_tau
SpinZ_tau

```

7. Read analysis results into a Pandas Dataframe with one row per simulation, containing parameters and observables:

```
[7]: obs = sim.get_obs()
```

```
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square
```

```
[8]: obs
```

```
[8]:
continuous ham_chem \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H...      0      0.0

ham_t ham_t2 ham_tperp \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H...      1.0      1.0      1.0

ham_u ham_u2 mz l1 l2 \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H...      4.0      4.0      1      6      6

... \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H...      ...

SpinXY_tauK \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... [[1.0321911697081605,
0.41267603482430026, 0.6...

SpinXY_tauK_err \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... [[0.22628722504550844,
0.37685459854025805, 0...

SpinXY_tauR \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... [[0.0575988003757706,
-0.10378742200171556, 0...

SpinXY_tauR_err \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... [[0.01248834666799511,
0.036871289378260834, 0...

SpinXY_tau_lattice \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... {'L1': [6.0, 0.0], 'L2':
[0.0, 6.0], 'a1': [1...

SpinZ_tauK \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... [[0.9040036362034092,
0.5628191896020613, 0.61...

SpinZ_tauK_err \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... [[0.1494786750311893,
0.06006336638595281, 0.0...

SpinZ_tauR \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... [[0.1013442053139478,
-0.11445552391592478, 0...

SpinZ_tauR_err \
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... [[0.06273438448505823,
0.056390780430637014, 0...

SpinZ_tau_lattice
/home/jonas/Programs/pyALF/Notebooks/ALF_data/H... {'L1': [6.0, 0.0], 'L2':
[0.0, 6.0], 'a1': [1...

[1 rows x 110 columns]
```

- The internal energy of the system (and its error) are accessed by:

```
[9]: obs.iloc[0][['Ener_scal0', 'Ener_scal0_err', 'Ener_scal_sign', 'Ener_scal_sign_err']]
```

```
[9]: Ener_scal0          -29.821914
Ener_scal0_err         0.13032
Ener_scal_sign          1.0
Ener_scal_sign_err      0.0
Name: /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square, dtype:
object
```

- The simulation can be resumed by calling `sim.run()` again, increasing the precision of results:

```
[10]: sim.run()
sim.analysis()
obs2 = sim.get_obs()
obs2.iloc[0][['Ener_scal0', 'Ener_scal0_err', 'Ener_scal_sign', 'Ener_scal_sign_err']]
```

Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square"
for Monte Carlo run.

Resuming previous run.

Run /home/jonas/Programs/ALF/Prog/ALF.out

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Analyzing /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square

/home/jonas/Programs/pyALF/Notebooks

Scalar observables:

Ener_scal

Kin_scal

Part_scal

Pot_scal

Histogram observables:

Equal time observables:

Den_eq

Green_eq

SpinT_eq

SpinXY_eq

SpinZ_eq

Time displaced observables:

Den_tau

Green_tau

SpinT_tau

SpinXY_tau

SpinZ_tau

/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square

```
[10]: Ener_scal0          -29.609245
Ener_scal0_err         0.136803
Ener_scal_sign          1.0
Ener_scal_sign_err      0.0
Name: /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square, dtype:
object
```

```
[11]: print("\nRunning again reduced the error from\n",
        obs.iloc[0][['Ener_scal0_err']], "\nto\n",
        obs2.iloc[0][['Ener_scal0_err']])
```

Running again reduced the error from

Ener_scal0_err 0.13032

Name: /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square, dtype:

object

to

Ener_scal0_err 0.136803

Name: /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square, dtype:

object

Note: To run a fresh simulation - instead of performing a refinement over previous run(s) - the Monte Carlo run directory should be deleted before rerunning.

1.1. Exercises

1. Check info/help commands such as `help(alf_src)`, `alf_src.get_ham_names()`, and `alf_src.get_default_params('Hubbard')`.
2. Rerun once again and check the new improvement in precision.
3. Look at a few other observables (`sim.analysis()` outputs the names of those available).
4. Change the lattice size by adding, e.g., `"L1": 4`, and `"L2": 1`, to the simulation parameters definitions of `sim` (step 2).

2. Trotter systematic error - Hubbard on the square lattice

In this example we use the [pyALF](#) interface to run ALF with the Mz choice of Hubbard-Stratonovich transformation (i.e., coupled to the z -component of the spin) on a 6×6 site square lattice, at $U/t = 4$ half-band filling, and inverse temperature $\beta t = 5$.

We carry out a systematic $\Delta\tau t$ extrapolation keeping $\Delta\tau t L_{\text{Trotter}} = 2$ constant. Recall that the formulation of the auxiliary field QMC approach is based on the symmetric Trotter decomposition

$$e^{-\Delta\tau(\hat{A}+\hat{B})} = e^{-\Delta\tau\hat{A}/2}e^{-\Delta\tau\hat{B}}e^{-\Delta\tau\hat{A}/2} + \mathcal{O}(\Delta\tau^3)$$

The overall error produced by this approximation is of the order $\Delta\tau^2$.

Bellow we go through the steps for performing this extrapolation: setting the simulation parameters, running it and analysing the data. A reference plot for this analyses is found in [ALF documentation](#), Sec. 2.3.2 (Symmetric Trotter decomposition).

-
1. Import `Simulation` class from the `py_alf` python module, which provides the interface with ALF, as well as mathematics and plotting packages:

```
[1]: from py_alf import ALF_source, Simulation # Interface with ALF
#
import numpy as np                        # Numerical library
from scipy.optimize import curve_fit      # Numerical library
import matplotlib.pyplot as plt           # Plotting library
alf_src = ALF_source(branch='ALF-2.3')    # Obtain ALF source code
```

Checking out branch ALF-2.3

Your branch is up to date with 'origin/ALF-2.3'.

Already on 'ALF-2.3'

2. Create instances of `Simulation`, specifying the necessary parameters, in particular the different $\Delta\tau$ values:

```
[2]: sims = []                                # Vector of Simulation instances
print('dtau values used:')
for dtau in [0.05, 0.1, 0.15]:              # Values of dtau
    print(dtau)
    sim = Simulation(
        alf_src,
        'Hubbard',
        {                                     # Hamiltonian
            'Model': 'Hubbard',               # Model and simulation parameters for each Simulation instance
            'Lattice_type': 'Square',         # Base model
            'L1': 6,                          # Lattice length in the first unit vector direction
            'L2': 6,                          # Lattice length in the second unit vector direction
            'Checkerboard': False,            # Whether checkerboard decomposition is used or not
            'Symm': True,                     # Whether symmetrization takes place
            'ham_T': 1.0,                     # Hopping parameter
            'ham_U': 4.0,                     # Hubbard interaction
            'ham_Tperp': 0.0,                 # For bilayer systems
            'beta': 5.0,                     # Inverse temperature
            'Ltau': 0,                       # '1' for time-displaced Green functions; '0' otherwise
        },
    )
```

```

    'NSweep': 100,          # Number of sweeps per bin
    'NBin': 10,            # Number of bins
    'Dtau': dtau,          # Only dtau varies between simulations, Ltrot=beta/Dtau
    'Mz': True,            # If true, sets the M_z-Hubbard model: Nf=2, N_sum=1,
                           # HS field couples to z-component of magnetization
)
sims.append(sim)

```

dtau values used:

```

0.05
0.1
0.15

```

3. Compile ALF, downloading it first if not found locally. This may take a few minutes:

```
[3]: sims[0].compile() # Compilation needs to be performed only once
```

```

Compiling ALF...
Cleaning up Prog/
Cleaning up Libraries/
Cleaning up Analysis/
Compiling Libraries

entanglement_mod.F90:35:2:

   35 | #warning "You are compiling entanglement without MPI. No results
      | 1~~~~~
Warning: #warning "You are compiling entanglement without MPI. No results
possible" [-Wcpp]
ar: creating modules_90.a
ar: creating libqrref.a

Compiling Analysis
Compiling Program
Parsing Hamiltonian parameters
filename: Hamiltonians/Hamiltonian_Kondo_smod.F90
filename: Hamiltonians/Hamiltonian_Hubbard_smod.F90
filename: Hamiltonians/Hamiltonian_Hubbard_Plain_Vanilla_smod.F90
filename: Hamiltonians/Hamiltonian_tV_smod.F90
filename: Hamiltonians/Hamiltonian_LRC_smod.F90
filename: Hamiltonians/Hamiltonian_Z2_Matter_smod.F90
Compiling program modules
Link program
Done.

```

4. Perform the simulations, as specified in each element of `sim`:

```
[4]: for i, sim in enumerate(sims):
      sim.run() # Perform the actual simulation in ALF
```

```

Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_
L1=6_L2=6_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.05_
Mz=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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This Program comes with ABSOLUTELY NO WARRANTY; for details see license.GPL
This is free software, and you are welcome to redistribute it under certain
conditions.
No initial configuration
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_
L1=6_L2=6_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.1_M
z=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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This is free software, and you are welcome to redistribute it under certain
conditions.
No initial configuration
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_
L1=6_L2=6_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.15_

```

Mz=True" for Monte Carlo run.
 Create new directory.
 Run /home/jonas/Programs/ALF/Prog/ALF.out
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 This Program comes with ABSOLUTELY NO WARRANTY; for details see license.GPL
 This is free software, and you are welcome to redistribute it under certain
 conditions.
 No initial configuration

5. Calculate the internal energies:

```
[5]: ener = np.empty((len(sims), 2))          # Matrix for storing energy values
      dtaus = np.empty((len(sims),))         # Matrix for Dtau values, for plotting
      for i, sim in enumerate(sims):
          print(sim.sim_dir)                 # Directory containing the simulation output
          sim.analysis()                     # Perform default analysis
          dtaus[i] = sim.sim_dict['Dtau']     # Store Dtau value
          obs = sim.get_obs()
          ener[i] = obs.iloc[0][['Ener_scal0', 'Ener_scal0_err']] # Store energy value
```

```
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_L2=6_Checkerbo
ard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.05_Mz=True
### Analyzing /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_
L2=6_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.05_Mz=Tr
ue ###
/home/jonas/Programs/pyALF/Notebooks
Scalar observables:
Ener_scal
Kin_scal
Part_scal
Pot_scal
Histogram observables:
Equal time observables:
Den_eq
Green_eq
SpinT_eq
SpinXY_eq
SpinZ_eq
Time displaced observables:
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_L2=6_Checkerbo
ard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.05_Mz=True
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_L2=6_Checkerbo
ard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.1_Mz=True
### Analyzing /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_
L2=6_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.1_Mz=Tru
e ###
/home/jonas/Programs/pyALF/Notebooks
Scalar observables:
Ener_scal
Kin_scal
Part_scal
Pot_scal
Histogram observables:
Equal time observables:
Den_eq
Green_eq
SpinT_eq
SpinXY_eq
SpinZ_eq
Time displaced observables:
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_L2=6_Checkerbo
ard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.1_Mz=True
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_L2=6_Checkerbo
ard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.15_Mz=True
### Analyzing /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_
L2=6_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.15_Mz=Tr
ue ###
/home/jonas/Programs/pyALF/Notebooks
Scalar observables:
Ener_scal
Kin_scal
Part_scal
Pot_scal
```

```

Histogram observables:
Equal time observables:
Den_eq
Green_eq
SpinT_eq
SpinXY_eq
SpinZ_eq
Time displaced observables:
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_Square_L1=6_L2=6_Checkerbo
ard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=5.0_Dtau=0.15_Mz=True

```

```
[6]: print('For Dtau values', dtaus, 'the measured energies are:\n', ener)
```

```

For Dtau values [0.05 0.1 0.15] the measured energies are:
[[-29.75742853  0.08560141]
 [-29.81231535  0.0437549 ]
 [-29.91469593  0.07072682]]

```

```

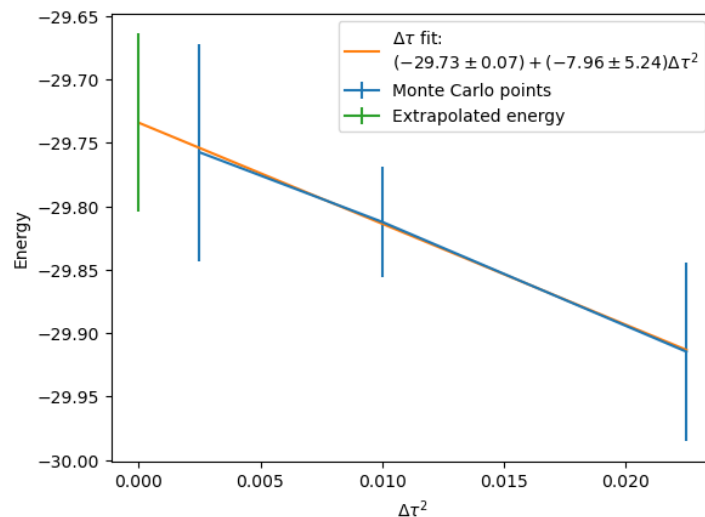
[7]: plt.xlabel(r'$\Delta\tau^2$')
plt.ylabel('Energy')
plt.errorbar(dtaus**2, ener[:, 0], ener[:, 1], label='Monte Carlo points')

def func(x, y0, a):
    return y0 + a*x**2
popt1, pcov = curve_fit(func, dtaus, ener[:, 0], sigma=ener[:, 1], absolute_sigma=True)
perr1 = np.sqrt(np.diag(pcov))

xs = np.linspace(0., dtaus.max())
plt.plot(xs**2, func(xs, *popt1), label=r'$\Delta\tau$ fit: '\n'
        f'${popt1[0]:.2f}\pm{perr1[0]:.2f})+({popt1[1]:.2f})\pm{perr1[1]:.2f})'
        r' \Delta\tau^2$'
        )
plt.errorbar(0, popt1[0], perr1[0], label='Extrapolated energy')
plt.legend()

```

```
[7]: <matplotlib.legend.Legend at 0x7f745b051930>
```



2.1. Exercises

1. Try out the four different combinations for **Checkerboard** and **Symm** settings in order to observe their effect on the output and run time. Reference: Sec. 2.3.2 - Symmetric Trotter decomposition - of the [ALF documentation](#), especially Fig. 2.

3. Testing against ED - Hubbard on a ring

In this example we use the [pyALF](#) interface to run ALF with the Mz choice of Hubbard Stratonovitch transformation on a four site ring, at $U/t = 4$ and inverse temperature $\beta t = 2$. For this set of parameters, the exact internal energy reads:

$$\left\langle -t \sum_{\langle i,j \rangle, \sigma} \hat{c}_{i,\sigma}^\dagger \hat{c}_{j,\sigma} + U \sum_{i=1}^N \hat{n}_{i,\uparrow} \hat{n}_{i,\downarrow} \right\rangle = -1.47261997t$$

To reproduce this result we will have to carry out a systematic $\Delta\tau t$ extrapolation keeping $\Delta\tau t L_{\text{Trotter}} = 2$ constant.

Recall that the formulation of the auxiliary field QMC approach is based on the Trotter decomposition

$$e^{-\Delta\tau(\hat{A}+\hat{B})} = e^{-\Delta\tau\hat{A}/2} e^{-\Delta\tau\hat{B}} e^{-\Delta\tau\hat{A}/2} + \mathcal{O}(\Delta\tau^3)$$

The overall error produced by this approximation is of the order $\Delta\tau^2$.

Bellow we go through the steps for performing this extrapolation: setting the simulation parameters, running it and analyzing the data.

1. Import Simulation class from the `py_alf` python module, which provides the interface with ALF, as well as mathematics and plotting packages:

```
[1]: from py_alf import Simulation, ALF_source # Interface with ALF
#
import numpy as np                        # Numerical library
from scipy.optimize import curve_fit      # Numerical library
import matplotlib.pyplot as plt          # Plotting library
alf_src = ALF_source(branch='ALF-2.3')   # Obtain ALF source code
```

Checking out branch ALF-2.3

Your branch is up to date with 'origin/ALF-2.3'.

Already on 'ALF-2.3'

2. Create instances of `Simulation`, specifying the necessary parameters, in particular the different $\Delta\tau$ values:

```
[2]: sims = []                                # Vector of Simulation instances
print('dtau values used:')
for dtau in [0.05, 0.1, 0.2]:                # Values of dtau
    print(dtau)
    sim = Simulation(
        alf_src,
        'Hubbard',                            # Hamiltonian
        {                                       # Model and simulation parameters for each Simulation instance
            'Lattice_type': 'N_leg_ladder',    # Lattice type
            'L1': 4,                          # Lattice length in the first unit vector direction
            'L2': 1,                          # Lattice length in the second unit vector direction
            'Checkerboard': False,             # Whether checkerboard decomposition is used or not
            'Symm': True,                     # Whether symmetrization takes place
            'ham_T': 1.0,                     # Hopping parameter
            'ham_U': 4.0,                     # Hubbard interaction
            'ham_Tperp': 0.0,                 # For bilayer systems
            'beta': 2.0,                     # Inverse temperature
            'Ltau': 0,                       # '1' for time-displaced Green functions; '0' otherwise
            'NSweep': 1000,                  # Number of sweeps per bin
            'NBin': 100,                    # Number of bins
            'Dtau': dtau,                   # Only dtau varies between simulations, Ltrot=beta/Dtau
            'Mz': True,                     # If true, sets the M_z-Hubbard model: Nf=2, N_sum=1,
        },                                    # HS field couples to z-component of magnetization
    )
    sims.append(sim)
```

dtau values used:

```
0.05
0.1
0.2
```

3. Compile ALF, downloading it first if not found locally. This may take a few minutes:

```
[3]: sims[0].compile() # Compilation needs to be performed only once
```

```
Compiling ALF...
Cleaning up Prog/
Cleaning up Libraries/
Cleaning up Analysis/
Compiling Libraries

entanglement_mod.F90:35:2:

    35 | #warning "You are compiling entanglement without MPI. No results
possible"
      | 1~~~~~
Warning: #warning "You are compiling entanglement without MPI. No results
possible" [-Wcpp]
ar: creating modules_90.a
ar: creating libqrref.a

Compiling Analysis
Compiling Program
Parsing Hamiltonian parameters
filename: Hamiltonians/Hamiltonian_Kondo_smod.F90
filename: Hamiltonians/Hamiltonian_Hubbard_smod.F90
filename: Hamiltonians/Hamiltonian_Hubbard_Plain_Vanilla_smod.F90
filename: Hamiltonians/Hamiltonian_tV_smod.F90
filename: Hamiltonians/Hamiltonian_LRC_smod.F90
filename: Hamiltonians/Hamiltonian_Z2_Matter_smod.F90
Compiling program modules
Link program
Done.
```

4. Perform the simulations, as specified in each element of `sim`:

```
[4]: for i, sim in enumerate(sims):
      sim.run() # Perform the actual simulation in ALF
```

```
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
adder_L1=4_L2=1_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau
=0.05_Mz=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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conditions.
No initial configuration
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
adder_L1=4_L2=1_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau
=0.1_Mz=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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conditions.
No initial configuration
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
adder_L1=4_L2=1_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau
=0.2_Mz=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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This is free software, and you are welcome to redistribute it under certain
conditions.
No initial configuration
```

5. Calculate the internal energies:

```
[5]: ener = np.empty((len(sims), 2))          # Matrix for storing energy values
      dtaus = np.empty((len(sims),))         # Matrix for Dtau values, for plotting
      for i, sim in enumerate(sims):
          print(sim.sim_dir)                  # Directory containing the simulation output
          sim.analysis()                      # Perform default analysis
          dtaus[i] = sim.sim_dict['Dtau']      # Store Dtau value
          obs = sim.get_obs()
          ener[i] = obs.iloc[0][['Ener_scal0', 'Ener_scal0_err']] # Store energy value
```

```
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder_L1=4_L2=1_Che
ckerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.05_Mz=True
### Analyzing /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder
_L1=4_L2=1_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.05
_Mz=True ###
/home/jonas/Programs/pyALF/Notebooks
Scalar observables:
Ener_scal
Kin_scal
Part_scal
Pot_scal
Histogram observables:
Equal time observables:
Den_eq
Green_eq
SpinT_eq
SpinXY_eq
SpinZ_eq
Time displaced observables:
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder_L1=4_L2=1_Che
ckerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.05_Mz=True
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder_L1=4_L2=1_Che
ckerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.1_Mz=True
### Analyzing /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder
_L1=4_L2=1_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.1_
Mz=True ###
/home/jonas/Programs/pyALF/Notebooks
Scalar observables:
Ener_scal
Kin_scal
Part_scal
Pot_scal
Histogram observables:
Equal time observables:
Den_eq
Green_eq
SpinT_eq
SpinXY_eq
SpinZ_eq
Time displaced observables:
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder_L1=4_L2=1_Che
ckerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.1_Mz=True
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder_L1=4_L2=1_Che
ckerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.2_Mz=True
### Analyzing /home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder
_L1=4_L2=1_Checkerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.2_
Mz=True ###
/home/jonas/Programs/pyALF/Notebooks
Scalar observables:
Ener_scal
Kin_scal
Part_scal
Pot_scal
Histogram observables:
Equal time observables:
Den_eq
Green_eq
SpinT_eq
SpinXY_eq
SpinZ_eq
Time displaced observables:
/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_ladder_L1=4_L2=1_Che
ckerboard=False_Symm=True_T=1.0_U=4.0_Tperp=0.0_beta=2.0_Dtau=0.2_Mz=True
```

```
[6]: print('For Dtau values', dtaus, 'the measured energies are:\n', ener)
```

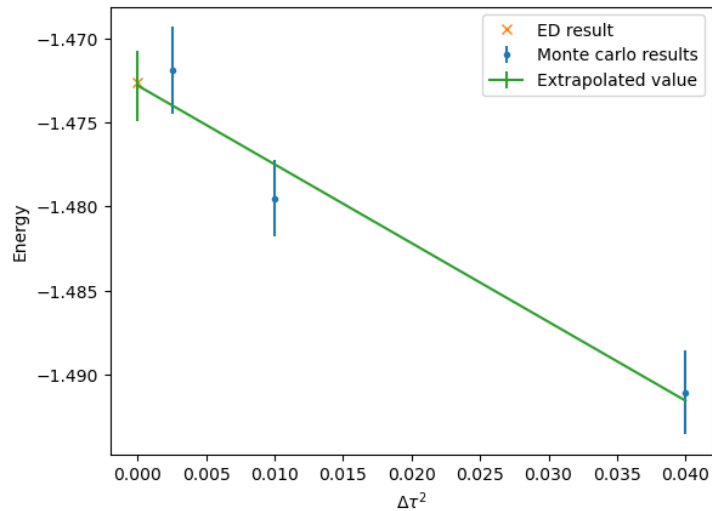
```
For Dtau values [0.05 0.1 0.2 ] the measured energies are:
[[-1.47188895  0.00256759]
 [-1.47953032  0.00227281]
 [-1.49103639  0.00250634]]
```

```
[8]: plt.figure()
plt.xlabel(r'$\Delta\tau^2$')
plt.ylabel('Energy')
plt.errorbar(dtaus**2, ener[:, 0], ener[:, 1], fmt='.', label="Monte carlo results")
plt.plot(0, -1.47261997, 'x', label='ED result')

def func(x, y0, a):
    return y0 + a*x**2
popt, pcov = curve_fit(func, dtaus, ener[:, 0], sigma=ener[:, 1], absolute_sigma=True)
perr = np.sqrt(np.diag(pcov))
xs = np.linspace(0., dtaus.max())
p = plt.plot(xs**2, func(xs, *popt))

plt.errorbar(0, popt[0], perr[0], label="Extrapolated value", color=p[0].get_color())
plt.legend()
```

```
[8]: <matplotlib.legend.Legend at 0x7ff978bcc0a0>
```



3.1. Exercises

1. Redo the extrapolation for different values of βt (e.g., for $\beta t = 1$, the internal energy is -0.62186692t, and for $\beta t = 4$, it is -1.90837196t).
2. Experiment with different settings for **Checkerboard** and **Symm**.

4. Projective algorithm

In this example we use the **pyALF** interface to run ALF's projective algorithm with the Mz choice of Hubbard Stratonovich transformation on a 4-site ring.

The projective approach is the method of choice if one is interested in ground-state properties. The starting point is a pair of trial wave functions, $|\Psi_{T,L/R}\rangle$, that are not orthogonal to the ground state $|\Psi_0\rangle$:

$$\langle \Psi_{T,L/R} | \Psi_0 \rangle \neq 0.$$

The ground-state expectation value of any observable $\hat{O}$ can then be computed by propagation along the imaginary time axis:

$$\frac{\langle \Psi_0 | \hat{O} | \Psi_0 \rangle}{\langle \Psi_0 | \Psi_0 \rangle} = \lim_{\theta \rightarrow \infty} \frac{\langle \Psi_{T,L} | e^{-\theta \hat{H}} e^{-(\beta-\tau) \hat{H}} \hat{O} e^{-\tau \hat{H}} e^{-\theta \hat{H}} | \Psi_{T,R} \rangle}{\langle \Psi_{T,L} | e^{-(2\theta+\beta) \hat{H}} | \Psi_{T,R} \rangle},$$

where β defines the imaginary time range where observables (time displaced and equal time) are measured and τ varies from 0 to β in the calculation of time-displace observables. For further details, see Sec. 3 of [ALF documentation](#).

1. Import Simulation class from the `py_alf` python module, which provides the interface with ALF, as well as numerical and plotting packages:

```
[1]: from py_alf import Simulation, ALF_source # Interface with ALF
#
import numpy as np                        # Numerical library
from scipy.optimize import curve_fit      # Numerical library
import matplotlib.pyplot as plt          # Plotting library
alf_src = ALF_source(branch='ALF-2.3')   # Obtain ALF source code
```

Checking out branch ALF-2.3

Your branch is up to date with 'origin/ALF-2.3'.

Already on 'ALF-2.3'

2. Create instances of Simulation, specifying the necessary parameters, in particular the Projector to True:

```
[24]: sims = []                                # Vector of Simulation instances
print('Theta values used:')
for theta in [0.5, 1, 1.5, 3, 5, 10]:        # Values of Theta
    print(theta)
    sim = Simulation(
        alf_src,
        'Hubbard',                            # Hamiltonian
        {                                     # Model and simulation parameters for each Simulation instance
            'Model': 'Hubbard',                # Base model
            'Lattice_type': 'N_leg_ladder',    # Lattice type
            'L1': 4,                           # Lattice length in the first unit vector direction
            'L2': 1,                           # Lattice length in the second unit vector direction
            'Checkerboard': False,              # Whether checkerboard decomposition is used or not
            'Symm': True,                      # Whether symmetrization takes place
            'Projector': True,                 # Whether to use the projective algorithm
            'Theta': theta,                    # Projector parameter
            'ham_T': 1.0,                      # Hopping parameter
            'ham_U': 4.0,                      # Hubbard interaction
            'ham_Tperp': 0.0,                  # For bilayer systems
            'beta': 0.5,                       # Inverse temperature
            'Ltau': 0,                         # '1' for time-displaced Green functions; '0' otherwise
            'NSweep': 600,                     # Number of sweeps per bin
            'NBin': 50,                       # Number of bins
            'Dtau': 0.05,                     # Ltrot=beta/Dtau
            'Mz': True,                       # If true, sets the M_z-Hubbard model: Nf=2, N_sum=1,
                                                # HS field couples to z-component of magnetization
        },
    )
    sims.append(sim)
```

Theta values used:

```
0.5
1
1.5
3
5
10
```

3. Compile ALF, this may take a few minutes:

```
[19]: sims[0].compile()                       # Compilation needs to be performed only once
```

```

Compiling ALF...
Cleaning up Prog/
Cleaning up Libraries/
Cleaning up Analysis/
Compiling Libraries

entanglement_mod.F90:35:2:

    35 | #warning "You are compiling entanglement without MPI. No results
possible"
      | 1~~~~~
Warning: #warning "You are compiling entanglement without MPI. No results
possible" [-Wcpp]
ar: creating modules_90.a
ar: creating libqrref.a

Compiling Analysis
Compiling Program
Parsing Hamiltonian parameters
filename: Hamiltonians/Hamiltonian_Kondo_smod.F90
filename: Hamiltonians/Hamiltonian_Hubbard_smod.F90
filename: Hamiltonians/Hamiltonian_Hubbard_Plain_Vanilla_smod.F90
filename: Hamiltonians/Hamiltonian_tV_smod.F90
filename: Hamiltonians/Hamiltonian_LRC_smod.F90
filename: Hamiltonians/Hamiltonian_Z2_Matter_smod.F90
Compiling program modules
Link program
Done.

```

4. Perform the simulations, as specified in each element of `sim`:

```
[25]: for i, sim in enumerate(sims):
      sim.run()                                # Perform the actual simulation in ALF
```

```

Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
adder_L1=4_L2=1_Checkerboard=False_Symm=True_Projector=True_Theta=0.5_T=1.0_U=4.
0_Tperp=0.0_beta=0.5_Dtau=0.05_Mz=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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conditions.
No initial configuration
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
adder_L1=4_L2=1_Checkerboard=False_Symm=True_Projector=True_Theta=1_T=1.0_U=4.0_
Tperp=0.0_beta=0.5_Dtau=0.05_Mz=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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conditions.
No initial configuration
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
adder_L1=4_L2=1_Checkerboard=False_Symm=True_Projector=True_Theta=1.5_T=1.0_U=4.
0_Tperp=0.0_beta=0.5_Dtau=0.05_Mz=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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conditions.
No initial configuration
Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
adder_L1=4_L2=1_Checkerboard=False_Symm=True_Projector=True_Theta=3_T=1.0_U=4.0_
Tperp=0.0_beta=0.5_Dtau=0.05_Mz=True" for Monte Carlo run.
Create new directory.
Run /home/jonas/Programs/ALF/Prog/ALF.out
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conditions.
No initial configuration

```

Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
 adder_L1=4_L2=1_Checkerboard=False_Symm=True_Projector=True_Theta=5_T=1.0_U=4.0_
 Tperp=0.0_beta=0.5_Dtau=0.05_Mz=True" for Monte Carlo run.

Create new directory.

Run /home/jonas/Programs/ALF/Prog/ALF.out

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No initial configuration

Prepare directory "/home/jonas/Programs/pyALF/Notebooks/ALF_data/Hubbard_N_leg_1
 adder_L1=4_L2=1_Checkerboard=False_Symm=True_Projector=True_Theta=10_T=1.0_U=4.0_
 Tperp=0.0_beta=0.5_Dtau=0.05_Mz=True" for Monte Carlo run.

Create new directory.

Run /home/jonas/Programs/ALF/Prog/ALF.out

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No initial configuration

5. Calculate the internal energies:

```
[26]: %%capture
ener = np.empty((len(sims), 2))           # Matrix for storing energy values
thetas = np.empty((len(sims),))          # Matrix for Thetas values, for plotting
for i, sim in enumerate(sims):
    print(sim.sim_dir)                    # Directory containing the simulation output
    sim.analysis()                         # Perform default analysis
    thetas[i] = sim.sim_dict['Theta']      # Store Theta value
    obs = sim.get_obs()
    ener[i] = obs.iloc[0][['Ener_scal0', 'Ener_scal0_err']] # Store energy value
```

Where the *cell magic* %%capture suppresses the output of `sim.analysis()`, which lists the data directories and observables.

```
[27]: print('For Theta values', thetas, 'the measured energies are:\n', ener)
```

For Theta values [0.5 1. 1.5 3. 5. 10.] the measured energies are:

```
[[[-2.05352132  0.00463096]
```

```
[-2.1085726  0.00680212]
```

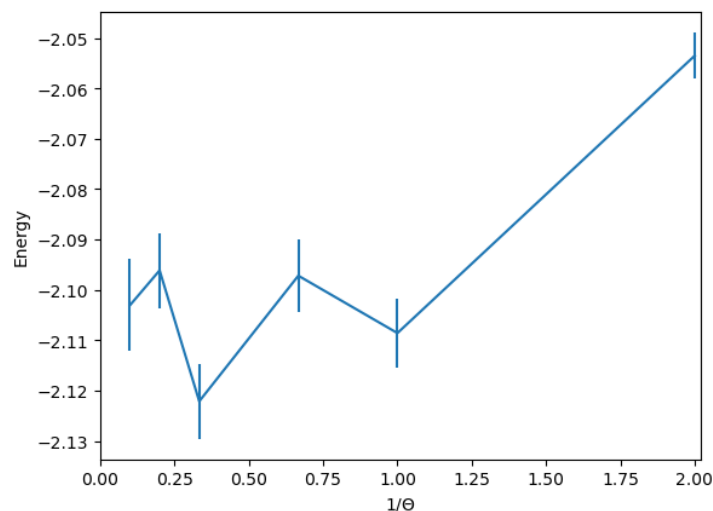
```
[-2.09718603  0.00725172]
```

```
[-2.12211566  0.0074775 ]
```

```
[-2.0961295  0.00745434]
```

```
[-2.10306699  0.0091302 ]]
```

```
[31]: plt.xlabel(r'$1/\Theta$')
plt.ylabel('Energy')
plt.errorbar(1/thetas, ener[:, 0], ener[:, 1]);
plt.xlim(0, 2.02);
```



4.1. Exercises

1. A ladder system consists of chains assembled one next to the other, for instance, setting $L_1=14$, $L_2=3$ defines a 3-leg ladder. It is a well-known result ([Dagotto and Rice, *Science* **271**, 5249, \(1996\) pp. 618](#)) that spin correlations in ladder systems decay as power laws (apart from logarithmic corrections) for odd-leg ladders, and exponentially for even-leg ladders. The paper presents numerical results for the Heisenberg model. How do these correlations behave for the Hubbard model at half-filling?

Part II. Getting your hands dirty - writing new code

A lot already comes implemented in ALF, but unavoidably, as one proceeds in their own investigations, a new model has to be implemented or a new observable defined – and for that, one must grapple with the package’s Fortran source code. This second part of the tutorial consists in a set of guided exercises that exemplify how to make basic additions to the code, taking as starting point the template-like, relatively self-contained submodule `Hamiltonian_Hubbard_Plain_Vanilla_smod.F90`, which is also a good display of ALF’s internal workings.

These worked-out exercises, together with ALF’s modularity, boosted by its Predefined Structures, should make getting your hands dirty less daunting than it may sound.

Downloading and using the code and tutorial

One can use the ALF package downloaded automatically by the Python script in the first part of this tutorial, or manually, by typing

```
git clone -b ALF-2.3 https://git.physik.uni-wuerzburg.de/ALF/ALF.git
```

in a shell. Similarly, to download the tutorial, including solutions, enter:

```
git clone https://git.physik.uni-wuerzburg.de/ALF/ALF_Tutorial.git
```

This document is found in the directory `Tutorial-ALF-2.3`.

The necessary environment variables and the directives for compiling the code are set by the script `configure.sh`: `source configure.sh GNU`, followed by the command `make`. Details and further options are described in the package’s documentation found in its repository.

A workflow you can adopt for solving the exercises – or indeed using ALF in general – is the following:

1. Compile the modified Hamiltonian module, for instance:
`make all`
2. Create a data directory with the content of `Start`:
`cp -r ./Scripts_and_Parameters_files/Start ./Run && cd ./Run/`
3. Run its executable, e.g., serially:
`$ALF_DIR/Prog/ALF.out`
4. Perform default analyses³:
`$ALF_DIR/Analysis/ana.out *`

The structure of the data files and details on the analysis output can be found in ALF’s documentation.

Exercise 1 – Dimensional crossover

Here we will modify the code so as to allow for different hopping matrix elements along the x and y directions of a square lattice.

1a) Modifying the hopping

To do so we start from the submodule `Hamiltonian_Hubbard_Plain_Vanilla_smod.F90`, which we here shorten to “Vanilla”, found in `$ALF_DIR/Prog/Hamiltonians/`, proceeding as follows:

- Add `Ham_Ty` to the `VAR_Hubbard_Plain_Vanilla` name space in the parameter file `parameters`.

³The `analysis.sh` bash script from earlier versions of ALF, run without arguments, is still available in the `Start` directory.

- Declare a new variable, `Ham_Ty`, in the module's specification (just search for the declaration of `Ham_T` in `Vanilla`).
- Add `Ham_Ty` to the `VAR_Hubbard_Plain_Vanilla` name space (`namelist`) declaration at the `Ham_Set` subroutine of `Vanilla`.
- Modify the hopping matrix in the subroutine `Ham_Hop` in `Vanilla`:

```

Do I = 1,Latt%N
  Ix = Latt%nnlist(I,1,0)
  Op_T(1,nf)%O(I, Ix) = cmplx(-Ham_T, 0.d0, kind(0.D0))
  Op_T(1,nf)%O(Ix, I) = cmplx(-Ham_T, 0.d0, kind(0.D0))
  If ( L2 > 1 ) then
    Iy = Latt%nnlist(I,0,1)
    !!!!!!! Modifications for Exercise 1a
    !Op_T(1,nf)%O(I, Iy) = cmplx(-Ham_T, 0.d0, kind(0.D0))
    !Op_T(1,nf)%O(Iy, I) = cmplx(-Ham_T, 0.d0, kind(0.D0))
    Op_T(1,nf)%O(I, Iy) = cmplx(-Ham_Ty, 0.d0, kind(0.D0))
    Op_T(1,nf)%O(Iy, I) = cmplx(-Ham_Ty, 0.d0, kind(0.D0))
    !!!!!!!
  endif
  Op_T(1,nf)%O(I, I) = cmplx(-Ham_chem, 0.d0, kind(0.D0))
  Op_T(1,nf)%P(i) = i
Enddo

```

It is a good idea as well to get the new simulation parameter written into the file `info`, also a change in `Ham_Set`.

In the directory `Solutions/Exercise_1` we have the modified and original submodules, as well as reference data and the necessary `Start` directory (remember to copy its contents to every new `Run` directory, and to have a different `Run` directory for each simulation).

As an application of this code, we can once again consider a ladder system (e.g. a 2-leg ladder with $L_1=14$ and $L_2=2$), for different values of `Ham_Ty`. The results you should obtain for the total spin correlation function (file `SpinT_eqJR`) are summarized in Fig. 1.

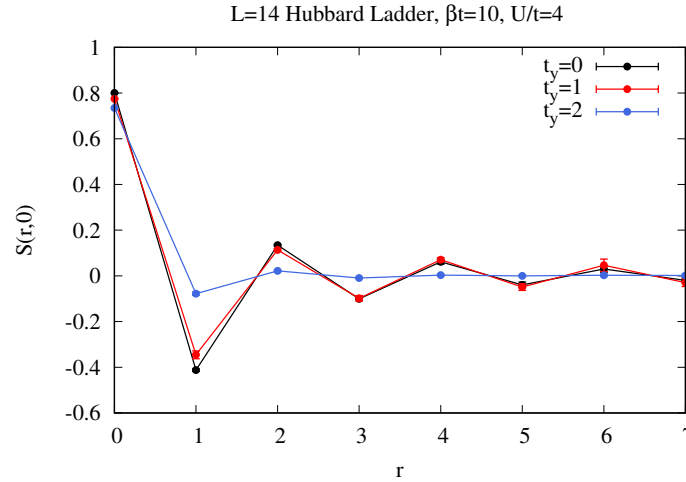


Figure 1: Spin correlation functions along one leg for the Hubbard ladder. As t_y grows the spin gap becomes large enough so as to detect the exponential decay of the spin correlation function on this small lattice size. The underlying physics of odd-even ladder systems is introduced in the article: Elbio Dagotto and T. M. Rice, “Surprises on the way from one- to two-dimensional quantum magnets: The ladder materials”, *Science* **271**, 5249 (1996) pp. 618–623.

1b) The SU(2) Hubbard-Stratonovich transformation

The SU(2) Hubbard-Stratonovich decomposition couples to the density and conserves spin rotational symmetry. Introduce into the module `Hamiltonian_Hubbard_smod.F90` and into the name space `VAR_Hubbard` the same changes done to the `Vanilla` module, described in the previous item, and enter `Mz=.F.` in the `parameters` file (in order to choose the $SU(N)$ Hubbard interaction) and compare results to those of the M_z decomposition above – especially with regard to numerical convergence.

Exercise 2 – Defining a new model: The one-dimensional t-V model

In this section, we will show what modifications have to be carried out for computing the physics of the one dimensional t-V model of spinless fermions.

$$\hat{H} = -t \sum_i \left(\hat{c}_i^\dagger \hat{c}_{i+a} + \hat{c}_{i+a}^\dagger \hat{c}_i \right) - \frac{V}{2} \sum_i \left(\hat{c}_i^\dagger \hat{c}_{i+a} + \hat{c}_{i+a}^\dagger \hat{c}_i \right)^2. \quad (1)$$

The above form is readily included in the ALF since the interaction is written in terms of a perfect square. Expanding the square yields (up to a constant) the desired model:

$$\hat{H} = -t \sum_i \left(\hat{c}_i^\dagger \hat{c}_{i+a} + \hat{c}_{i+a}^\dagger \hat{c}_i \right) + V \sum_i (\hat{n}_i - 1/2) (\hat{n}_{i+a} - 1/2). \quad (2)$$

Note that the t-V model is already implemented in ALF in the module `Hamiltonian_tV_smod.F90` found in `$ALF_DIR/Prog/Hamiltonians/`. While it can be used for checking your own results⁴, you are not supposed to reproduce that implementation – which is more general and makes use of predefined structures – but instead to write a simpler one, based on the module `Hamiltonian_Hubbard_Plain_Vanilla_smod.F90` as detailed below.

2a) Define new model

In the directory `Solutions/Exercise_2` we have duplicated the ALF and commented the changes that have to be carried out to the file `Hamiltonian_Hubbard_Plain_Vanilla_smod.F90`, which we here shorten to “`Vanilla`”, found in `$ALF_DIR/Prog/Hamiltonians/`. The following are the essential steps to be carried out:

- Add the `VAR_t_V` name space in the file `parameters` and set the necessary variables – or simply rename the `VAR_Hubbard_Plain_Vanilla` name space to `VAR_t_V` and, within it, `Ham_U` to `Ham_Vint`. (Ignore the name space `VAR_tV`, which is used by the general implementation mentioned above.)
- Declare a new variable, `Ham_Vint`, in `Vanilla`’s specification.
- Add the `VAR_t_V` name space (`namelist`) declaration at the `Ham_Set` subroutine of `Vanilla`, containing the same variables the name space contains in `parameters`, and read it in.
- Still in the `Ham_set` subroutine of `Vanilla`: set `NF=1`, since we are working with spinless fermions; change the output to the info file.
- In the `Ham_V` subroutine you have to add the new interaction. For a given bond at a given time slice, we need to decouple the interaction:

$$e^{\Delta\tau \frac{V}{2} (\hat{c}_i^\dagger \hat{c}_{i+a} + \hat{c}_{i+a}^\dagger \hat{c}_i)^2} = \sum_{l=\pm 1, \pm 2} \gamma_l e^{\sqrt{\Delta\tau \frac{V}{2}} \eta_l (\hat{c}_i^\dagger \hat{c}_{i+a} + \hat{c}_{i+a}^\dagger \hat{c}_i)} = \sum_{l=\pm 1, \pm 2} \gamma_l e^{g \eta_l (\hat{c}_i^\dagger, \hat{c}_{i+a}^\dagger) O(\hat{c}_i, \hat{c}_{i+a})^T}. \quad (3)$$

Here is how this translates in the code (the new integer variable, `i2`, should be declared):

```
Allocate(Op_V(Ndim,N_FL))
do nf = 1,N_FL
  do i = 1, Ndim
    call Op_make(Op_V(i,nf),2)
  enddo
```

⁴A short simulation of the t-V model can be conveniently run using its Jupyter notebook available in pyALF.

```

enddo
Do i = 1, Ndim ! Runs over bonds = # of lattice sites in one-dimension.
  i2
    = Latt%nnlist(i,1,0)
  Op_V(i,nf)%P(1) = i
  Op_V(i,nf)%P(2) = i2
  Op_V(i,nf)%O(1,2) = cmplx(1.d0 ,0.d0, kind(0.d0))
  Op_V(i,nf)%O(2,1) = cmplx(1.d0 ,0.d0, kind(0.d0))
  Op_V(i,nf)%g = sqrt(cmplx(Dtau*Ham_Vint/2.d0, 0.d0, kind(0.d0)))
  Op_V(i,nf)%alpha = cmplx(0d0 ,0.d0, kind(0.d0))
  Op_V(i,nf)%type = 2
  Call Op_set( Op_V(i,nf) )
enddo

```

- Finally, you have to update the **Obser** and **ObserT** routines for the calculation of equal- and time-displaced correlations. For the **t_V** model you can essentially use the same observables as for the **Hubbard_SU(2)** model in 1D – a step which requires a number of changes with respect to the **Vanilla** base, such as:

```

!!!! Modifications for Exercise 2
!Zpot = Zpot*ham_U ! Vanilla
Zpot = Zpot*Ham_Vint ! t-V
!!!!

```

and

```

!Zrho = Zrho + Grc(i,i,1) + Grc(i,i,2) ! Vanilla
Zrho = Zrho + Grc(i,i,1) ! t-V

```

with the observables being coded in the routine **Obser** as

```

Z = cmplx(dble(N_SUN), 0.d0, kind(0.d0))
Do I1 = 1,Ndim
  I = I1
  no_I = 1
  Do J1 = 1,Ndim
    J = J1
    no_J = 1
    imj = latt%imj(I,J)
    Obs_eq(1)%Obs_Latt(imj,1,no_I,no_J) = Obs_eq(1)%Obs_Latt(imj,1,no_I,no_J) + &
& Z * GRC(I1,J1,1) * ZP*ZS ! Green
    Obs_eq(2)%Obs_Latt(imj,1,no_I,no_J) = Obs_eq(2)%Obs_Latt(imj,1,no_I,no_J) + &
& Z * GRC(I1,J1,1) * GR(I1,J1,1) * ZP*ZS ! SpinZ
    Obs_eq(3)%Obs_Latt(imj,1,no_I,no_J) = Obs_eq(3)%Obs_Latt(imj,1,no_I,no_J) + &
& ( GRC(I1,I1,1) * GRC(J1,J1,1) * Z + &
& GRC(I1,J1,1) * GR(I1,J1,1) ) * Z * ZP*ZS ! Den
  enddo
  Obs_eq(3)%Obs_Latt0(no_I) = Obs_eq(3)%Obs_Latt0(no_I) + Z * GRC(I1,I1,1) * ZP * ZS
enddo

```

among other changes - with similar ones in the **ObserT** routine.

All necessary changes are implemented and clearly indicated in the solution provided in **Solutions/Exercise_2/Hamiltonian_Hubbard_Plain_Vanilla_smod-Exercise_2.F90**.

In the directory **Solutions/Exercise_2** we have the modified and original submodules, as well as reference data and the necessary **Start** directory (remember to copy its contents to every new **Run** directory, and to have a different **Run** directory for each simulation).

2b) Phase transition

You can now run the code for various values of V/t . A Jordan-Wigner transformation will map the $\mathbf{t}_V$ model onto the XXZ chain:

$$\hat{H} = J_{xx} \sum_i \hat{S}_i^x \hat{S}_{i+a}^x + \hat{S}_i^y \hat{S}_{i+a}^y + J_{zz} \sum_i \hat{S}_i^z \hat{S}_{i+a}^z, \quad (4)$$

with $J_{zz} = V$ and $J_{xx} = 2t$. Hence, when $V/t = 2$ we reproduce the Heisenberg model. For $V/t > 2$ the model is in the Ising regime with long-range charge density wave order and is an insulator. In the regime $-2 < V/t < 2$ the model is metallic and corresponds to a Luttinger liquid. Finally, at $V/t < -2$ phase separation between hole rich and electron rich phases occur. Fig. 2 shows typical results for the density-density correlation function (file `Den_eqJR`).

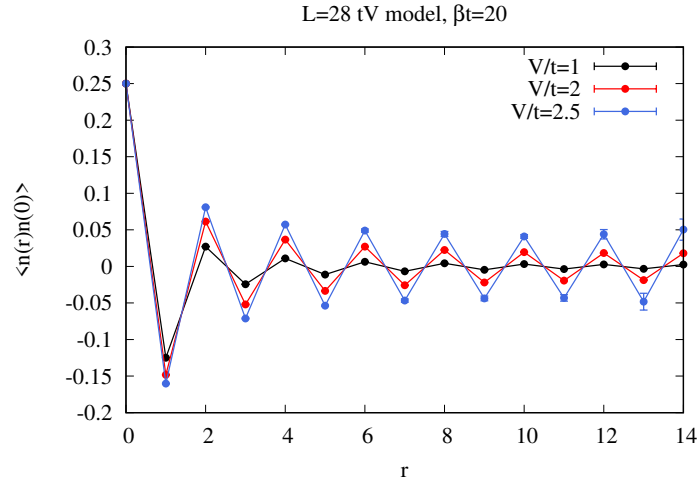


Figure 2: Density-Density correlation functions of the t - V model. In the Luttinger liquid phase, $-2 < V/t < 2$, it is known that the density-density correlations decay as $\langle n(r)n(0) \rangle \propto \cos(\pi r) r^{-(1+K_\rho)}$ with $(1 + K_\rho)^{-1} = \frac{1}{2} + \frac{1}{\pi} \arcsin\left(\frac{V}{2|t|}\right)$ (A. Luther and I. Peschel, Calculation of critical exponents in two dimensions from quantum field theory in one dimension, Phys. Rev. B **12** (1975), pp. 3908–3917.) The interested reader can try to reproduce this result.

Exercise 3 – Adding a new observable

This exercise illustrates the modifications that are required to implement a new observable, the correlation function of the bond-hopping in the 1-dimensional Hubbard chain. This observable is interesting in many different setups. For example, the model studied here exhibits an emergent $SO(4)$ symmetry that relates the anti-ferromagnetic order parameter and the bond dimerization (I. Affleck, PRL **55** (1985) pp. 1355; I. Affleck and F. D. M. Haldane, PRB **36** (1987) pp. 5291). Another example where this quantity is useful to investigate is the 1-dimensional Su-Schrieffer-Heger model describing an electron-phonon system.

3a) Applying Wick's theorem

Here the task is to define a new equal-time observable, the kinetic energy correlation, given by

$$\langle \hat{O}_{i,\delta} \hat{O}_{j,\delta'} \rangle - \langle \hat{O}_{i,\delta} \rangle \langle \hat{O}_{j,\delta'} \rangle = S_O(i - j, \delta, \delta') \quad (5)$$

where i, j refer to the unit cells and δ encodes the bond label. Since we are working in 1D, there is only one bond per unit cell and $\delta = ax$ such that

$$\hat{O}_{i,x} = \sum_{\sigma} \left(\hat{c}_{i,\sigma}^\dagger \hat{c}_{i+ax,\sigma} + \text{H.c.} \right). \quad (6)$$

Note that the first term of Eq. 5 is of the generic form $\sum_{\sigma,\sigma'} \langle \hat{c}_{i_1,\sigma}^\dagger \hat{c}_{i_2,\sigma} \hat{c}_{j_1,\sigma'}^\dagger \hat{c}_{j_2,\sigma'} \rangle$. This expectation value can be readily decomposed into single-particle Green functions by using Wick's theorem. It can be applied for a fixed field configuration Φ since the Hamiltonian $\hat{H}(\Phi)$ is then bi-linear in the fermion operators.

$$\sum_{\sigma,\sigma'} \langle \hat{c}_{i_1,\sigma}^\dagger \hat{c}_{i_2,\sigma} \hat{c}_{j_1,\sigma'}^\dagger \hat{c}_{j_2,\sigma'} \rangle_\Phi = \sum_{\sigma,\sigma'} \left(\langle \hat{c}_{i_1,\sigma}^\dagger \hat{c}_{i_2,\sigma} \rangle_\Phi \langle \hat{c}_{j_1,\sigma'}^\dagger \hat{c}_{j_2,\sigma'} \rangle_\Phi + \langle \hat{c}_{i_1,\sigma}^\dagger \hat{c}_{j_2,\sigma'} \rangle_\Phi \langle \hat{c}_{i_2,\sigma} \hat{c}_{j_1,\sigma'}^\dagger \rangle_\Phi \right) \quad (7)$$

The second term of vanishes for $\sigma \neq \sigma'$ due to flavor symmetry (Mz-decoupling used here in the vanilla version) or due to the $SU(2)$ symmetry (density decoupling available in the generic implementation of Hubbard model). The single-particle Green functions are provided in the **Obser** routine, where all equal-time observables are measured, as

$$GRC(i, j, \sigma) = \langle \hat{c}_{i,\sigma}^\dagger \hat{c}_{j,\sigma} \rangle_\Phi \quad (8)$$

$$GR(i, j, \sigma) = \langle \hat{c}_{i,\sigma} \hat{c}_{j,\sigma}^\dagger \rangle_\Phi. \quad (9)$$

3b) Necessary code modifications

In the directory `Solutions/Exercise_3` we have the modified and original submodule `Hamiltonian_Hubbard_Plain_Vanilla_smod.F90` found in `$ALF_DIR/Prog/Hamiltonians` (which we here shorten to “Vanilla”), as well as reference data and the necessary **Start** directory (remember to copy its contents to every new **Run** directory, and to have a different **Run** directory for each simulation). The following are the essential steps to be carried out:

- Introduce the new observable and allocate the memory required to store the measurements. This is done in the subroutine `Alloc_obs(Ltau)` by increasing the length of the array `Obs_eq` appropriately and adding a new case to specify the filename in which the results are stored on disc. (You might want to revisit this section later on to add the time-displaced version of the correlation function by changing `Obs_tau` in the same fashion.)
- The actual measurements are taken in the subroutine `Obser(GR,Phase,Ntau)`. While `GR` is passed to the subroutine, the first lines of code already implement the construct `GRC = 1 - GRT`. (This section does not have to be modified, but it is useful to keep this in mind for future reference when you implement a new model from scratch.)
- The measurement of an equal-time correlation function consists of two separate parts: the connected one given by the first term in Eq. 5, and the background, given by the second term.
- Implement the measurement of the connected part, stored as `Obs_eq(6)` in this example, using the Wick decomposition sketched above.
- Keep in mind that the background $\sum_i \langle \hat{O}_{i,\delta} \rangle \neq 0$ is non-vanishing and has to be measured separately (you can compare with the density correlation function), and is stored in `Obs_eq(6)%Obs_Latt0(1)`.
- The analysis tool will then automatically combine both contributions and evaluate Eq. 5 using the jackknife method to estimate the mean and error or the correlation function.

The 1-D Hubbard exhibits an emergent $SO(4)$ symmetry:

$$\langle \bar{S}(r) S(0) \rangle \sim \frac{(-1)^r}{r} \ln^d(r) \quad (10)$$

$$\langle \hat{O}_{r,x} \hat{O}_{0,x} \rangle - \langle \hat{O}_{r,x} \rangle \langle \hat{O}_{0,x} \rangle \sim \frac{(-1)^r}{r} \ln^\beta(r) \quad (11)$$

where $d = 1/2$ and $\beta = -3/2$ (T. Sato, M. Hohenadler et.al, Phys. Rev. B **104** (2021) L161105) and this exercise provides all the tools required to study it. Beware of the large system sizes, and therefore long run times, that are required to extract the logarithmic scaling corrections. More details are discussed in the appendix of above reference.

Finally, it is straightforward to implement the time-displaced version of this correlation function, following essentially the same steps as described above. The observable is now stored in `Obs_tau`, the measurements are taken in `ObserT`, and you can find the implementation in the solution to this exercise as well.